

September 12, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript, "Physics-Exact Credit Assignment for Multi-Agent Flutter Suppression," for consideration as a Regular Paper in Array. This submission is a transfer, via Elsevier's Article Transfer Service, of manuscript JSV-D-26-02695 from the Journal of Sound and Vibration, which recommended Array among its suggested destinations after declining the manuscript as outside its scope.

This manuscript has been submitted to three prior venues, each of which returned it as outside their scope, for three different reasons. The Journal of Guidance, Control, and Dynamics found it insufficiently narrow to guidance, navigation, and flight control in the sense that journal covers. AIAA Journal returned it as outside its scope of fundamentals and methods contributions because of its applied nature. The Journal of Sound and Vibration, most recently, found it outside its scope of sound- and vibration-phenomena contributions and suggested a journal on control or a specialist aircraft journal instead. Read together, we believe these three verdicts agree on what this manuscript actually is: a multi-agent reinforcement-learning method, validated on a physically grounded aeroelastic control problem, rather than a contribution to any of those three fields' own core subject matter. We believe Array – whose scope explicitly includes Artificial Intelligence, Machine Learning and Robotics, and Control and Signal Processing, alongside Interdisciplinary Applications – is the venue where that is the point of the paper rather than a categorisation difficulty.

The paper studies active flutter suppression on a wing with several independently actuated trailing-edge surfaces, each commanded by its own reinforcement-learning agent from local sensing alone – the setting a physically distributed avionics architecture actually presents, as opposed to the single-agent, single-effector, globally-observed setting the existing flutter-suppression literature uses throughout. Posing the problem this way exposes a credit-assignment question with a physical answer: for this class of energy-conserving structural plant, we show that the per-agent credit multi-agent reinforcement learning normally has to estimate is instead available in closed form directly from the equations of motion, and we validate it against a naive undecomposed reward and an adapted value-decomposition baseline (Holm-Bonferroni corrected across the full comparison family), against a six-rung classical baseline ladder (including a fully decentralised LQG synthesised from the same sensing the learned agents have), and against an unsteady vortex-lattice solver for the underlying plant. A second, more general finding follows from the same study: on this problem, the initial policy-exploration scale dominated every algorithmic choice we tried, by a factor of two, and is computable in advance from a classical design – a diagnostic we believe generalises to other physical control tasks with a narrow useful actuation band. We report two negative results (learned communication did not help; a shaping-based refinement did not survive correction for multiple comparisons once compared against the full family of alternatives) at the same statistical standard as the positive ones.

The manuscript's author list is restored here to the full collaborative team for this project; the three prior submissions listed the corresponding author alone. We disclose this change transparently and will complete Array's Authorship Change Request process, with written confirmation from all four authors, alongside this submission.

Other than the three prior submissions disclosed above, this manuscript, in whole or in substantial part, has not previously been submitted elsewhere. We will provide any further disclosure the editorial system requests. We declare no conflicts of interest, and no funding was received for this

work. Data and code needed to reproduce every result are cited in the manuscript's Data and Code Availability section.

Thank you for your consideration. We look forward to your response.

Sincerely,

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